

# Orbital Localization, Clifford Frames, and Fiedler Ordering in DMRG Benchmarks for $N_2$ and Linear $H_8$

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## Abstract

We compare fermionic spin-adapted and qubit-space density matrix renormalization group (DMRG) calculations for correlated  $N_2$  and linear  $H_8$  in the STO-3G basis. The qubit Hamiltonians are considered before and after Clifford transformations determined by two approximate-symmetry selection procedures: Hamiltonian coefficient thresholding (HCT) and Beam search. Four spatial-orbital bases—canonical, split Pipek–Mezey, split Pipek–Mezey restricted by point-group irrep, and CISD natural orbitals—are tested with their original ordering and with a Fiedler ordering derived from CISD mutual information. Clifford transformation substantially reduces the qubit-MPS bond dimension needed for chemical accuracy. For  $N_2$ , the best transformed calculations require bond dimension 8, compared with 39 for the canonical raw-qubit Hamiltonian. For linear  $H_8$ , split Pipek–Mezey localization and Fiedler ordering act cooperatively: the required bond dimensions are 8, 57, 30, and 19 for fermionic, raw-qubit, HCT, and Beam calculations, respectively. The specialized fermionic  $SU(2)$  implementation remains faster than all qubit calculations. The results also show that a smaller state bond dimension need not imply a shorter calculation when the Clifford transformation substantially increases the MPO bond dimension.

## 1 Scope of the comparison

For each molecular system and orbital basis, we benchmark four Hamiltonian frames:

1. a fermionic quantum-chemistry Hamiltonian treated with  $SU(2)$ -adapted DMRG;
2. the untransformed Jordan–Wigner qubit Hamiltonian;
3. the qubit Hamiltonian transformed by a Clifford circuit synthesized from HCT approximate symmetries; and
4. the qubit Hamiltonian transformed by a Clifford circuit synthesized from Beam-search approximate symmetries.

Every frame is tested in its standard site order and in a Fiedler order obtained from the corresponding CISD state. The primary outcome is the smallest tested maximum MPS bond dimension that simultaneously reaches chemical accuracy and satisfies the DMRG sweep-energy convergence criterion. The fermionic and qubit bond dimensions characterize different tensor-network representations and should not be interpreted as equal computational resources: the fermionic calculation uses four-state spatial-orbital sites and  $SU(2)$  block sparsity, whereas the qubit calculation uses two-state spin-orbital sites without particle-number or spin adaptation.

## 2 Computational methods

### 2.1 Molecules and electronic-structure calculations

All calculations use the STO-3G basis and retain all spatial orbitals and electrons.  $N_2$  is evaluated at an N–N distance of 1.4 Å; it contains 14 electrons in 10 spatial orbitals, corresponding to 20 qubits.  $H_8$  is a linear, equally spaced chain with adjacent H–H separation 2.0 Å; it contains 8 electrons in 8 spatial orbitals, corresponding to 16 qubits. The chains are centered on the Cartesian  $z$  axis. Restricted Hartree–Fock (RHF) calculations are performed in the Abelian  $D_{2h}$  subgroup with an SCF energy tolerance of  $10^{-12}$  and a maximum of 100 cycles. Spin orbitals are interleaved as  $(\alpha_0, \beta_0, \alpha_1, \beta_1, \dots)$  before the Jordan–Wigner transformation.

For every orbital basis, the one- and two-electron integrals, CISD state, and FCI reference energy are recomputed. FCI invariance under the orbital rotations is verified to  $10^{-8}$  Ha, and the FCI state is required to satisfy  $\langle \hat{S}^2 \rangle < 10^{-7}$ . The canonical reference energies are listed in Table 1. CISD is not invariant to the occupied–virtual mixing used to define the natural orbitals; its energy is therefore recomputed rather than copied from the canonical calculation.

Table 1: Electronic-structure dimensions and canonical-orbital reference energies. All energies are in Hartree.

| System         | $N_e$ | $N_{\text{orb}}$ | $N_q$ | $E_{\text{RHF}}$ | $E_{\text{CISD}}$ | $E_{\text{FCI}}$ |
|----------------|-------|------------------|-------|------------------|-------------------|------------------|
| N <sub>2</sub> | 14    | 10               | 20    | -107.3578154453  | -107.5832737773   | -107.6231741781  |
| H <sub>8</sub> | 8     | 8                | 16    | -3.1614329658    | -3.5542953056     | -3.7966934506    |

## 2.2 Orbital bases

Four spatial-orbital bases are compared.

**Canonical:** symmetry-adapted RHF molecular orbitals.

**Split PM:** Pipek–Mezey localization is applied separately to the canonical occupied and virtual subspaces. The occupied–virtual partition is not mixed.

**Split PM, irrep-restricted:** each occupied or virtual block is divided further according to its canonical  $D_{2h}$  irrep, and Pipek–Mezey localization is performed independently in every nontrivial block.

**Natural:** the spin-summed spatial one-particle density matrix of the canonical CISD state is diagonalized globally. The resulting orbitals are sorted by decreasing occupation. This transformation may mix canonical occupied and virtual orbitals; the CISD state is consequently recomputed in the natural-orbital basis.

The Pipek–Mezey convergence tolerance is  $10^{-10}$  with at most 1000 localization cycles. Orbital signs are fixed deterministically from the largest-magnitude AO coefficient. Orthonormality is verified to  $10^{-9}$ ; for the split localizations, preservation of the occupied projector is verified to  $10^{-8}$ .

## 2.3 Pauli Hamiltonians and approximate symmetries

The fermionic Hamiltonian is mapped directly to a packed Pauli representation with the repository’s streamed Jordan–Wigner implementation. Equal Pauli words are combined, and combined coefficients with magnitude at most  $10^{-12}$  are discarded. A Pauli word is stored as its binary  $X$  and  $Z$  masks, avoiding construction of an intermediate `OpenFermion QubitOperator`. The stream retains the first-occurrence order induced by the fermionic-term order and its Jordan–Wigner expansion. No subsequent sorting by coefficient magnitude, Pauli weight, or spatial support is used.

For a Pauli product  $S$ , both symmetry searches use the CISD squared-commutator cost

$$c(S) = \langle \Psi_{\text{CISD}} | [H, S]^\dagger [H, S] | \Psi_{\text{CISD}} \rangle. \quad (1)$$

Each search returns  $N_q$  linearly independent, mutually commuting Pauli generators. HCT progressively removes Hamiltonian terms by increasing a coefficient threshold over the distinct coefficient magnitudes above  $10^{-5}$ . At each threshold, the symplectic null space is reduced to a maximal commuting subspace, with candidate generators ordered first by Eq. (1), then by Pauli weight, and finally by their packed binary representation. Previously selected generators are retained as the threshold increases.

The Beam candidate pool contains the HCT generators, generators obtained from at most 256 Hamiltonian terms, and pairwise products seeded by the 50 leading Hamiltonian terms. The pool is sorted by increasing singleton cost and capped at 1000 candidates; no maximum Pauli weight is imposed. Beam search has width 16 and selects a full-rank commuting set by maximizing the negative sum of the singleton costs, equivalently minimizing  $\sum_S c(S)$ . The configured heavy-core fraction is 0.95. The resulting total costs are reported in Table 2; Fiedler reordering does not alter the selected symmetry set or its cost.

Table 2: Total CISD squared-commutator cost  $\sum_S c(S)$  of the selected full-rank symmetry sets.

| System         | Orbital basis              | HCT   | Beam  |
|----------------|----------------------------|-------|-------|
| N <sub>2</sub> | Canonical                  | 2.724 | 2.090 |
|                | Split PM                   | 3.602 | 2.779 |
|                | Split PM, irrep-restricted | 2.691 | 2.091 |
|                | Natural                    | 2.643 | 2.070 |
| H <sub>8</sub> | Canonical                  | 6.224 | 3.409 |
|                | Split PM                   | 1.253 | 0.617 |
|                | Split PM, irrep-restricted | 4.425 | 2.987 |
|                | Natural                    | 5.968 | 3.514 |

For each selected set, a Clifford circuit maps the generators to positive single-qubit  $Z$  operators. The same circuit is applied to the Hamiltonian and the CISD state, so the spectrum and the warm-start expectation value are preserved. Clifford action on the sparse state is evaluated exactly by bit-string and coefficient updates before conversion to an MPS; no determinant truncation is applied during the transformation.

## 2.4 Fiedler ordering

Fiedler orderings are derived from the CISD state separately for every Hamiltonian frame. For raw-qubit calculations, the graph vertices are individual spin-orbital qubits. For the fermionic calculation, each vertex is a four-state spatial orbital containing its paired  $\alpha$  and  $\beta$  modes. For HCT and Beam frames, the CISD state is first transformed by the corresponding Clifford circuit and the mutual information is then evaluated in that frame. One- and two-site reduced density matrices are contracted from the sparse state, entropies are evaluated in base 2, and the standard mutual information convention is used. Edges below  $10^{-12}$  are omitted. Sites are ordered by the Fiedler vector of the weighted graph Laplacian, with index tie-breaking and disconnected components ordered by total edge weight. The resulting permutation is applied identically to the Hamiltonian, symmetry operators, and CISD state.

## 2.5 MPO and MPS construction

Fermionic MPOs are built directly from the spatial one- and two-electron integrals with the SU(2)-adapted quantum-chemistry MPO builder in `pyblock2`; no nontrivial point-group labels are supplied to Block2. Qubit MPOs are built from the Pauli stream with Block2’s `FastBlockedSumBipartite` algorithm, using a block parameter of 10 and an MPO compression cutoff of  $10^{-10}$ , without an explicit MPO bond cap. Pauli products are supplied in their deterministic stream order with symbolic reordering disabled.

The sparse CISD state is converted to an MPS by adding determinants in decreasing coefficient magnitude in batches of 32 and compressing with an SVD cutoff of  $10^{-13}$ . The temporary bond cap during construction and Clifford transformation equals the largest bond dimension in the corresponding DMRG search. The constructed MPS is normalized, and its expectation value with the final MPO must agree with the independently computed CISD energy within  $10^{-7}$  Ha when the accumulated construction loss is below  $10^{-12}$ .

## 2.6 DMRG convergence search

All fermionic and qubit-space DMRG calculations use two-site updates in `pyblock2`. Each tested maximum bond dimension  $\chi$  starts from a fresh copy of the same CISD warm-start MPS rather than from the result at a neighboring bond dimension. DMRG is run for at most 100 sweeps with fixed maximum bond dimension  $\chi$ . The noise schedule is

$$(10^{-4}, 10^{-4}, 10^{-5}, 10^{-5}, 10^{-6}, 10^{-6}, 0, \dots). \quad (2)$$

The local Davidson threshold is  $10^{-10}$ , with at most 50 Davidson iterations per local optimization. Sweep convergence requires that the absolute change between the final two sweep energies be below  $10^{-6}$  Ha after the noise has been removed. A bond dimension is accepted only if this condition and chemical accuracy both hold:

$$|E_{\text{DMRG}}(\chi) - E_{\text{FCI}}| \leq 1.6 \times 10^{-3} \text{ Ha}. \quad (3)$$

An integer bisection search locates the smallest accepted  $\chi$  in  $1 \leq \chi \leq 200$  for  $\text{N}_2$  and  $1 \leq \chi \leq 300$  for  $\text{H}_8$ . The  $\text{N}_2$  run used the midpoint-first implementation; the completed  $\text{H}_8$  data used the earlier endpoint-bracketed implementation. In every successful frame the immediately lower integer was evaluated and failed, and no evaluated frame showed a successful smaller bond dimension followed by a failed larger one. This establishes the reported minimum within the individual searches, subject to the usual possibility of DMRG convergence to different local stationary states.

The runtime reported in the result tables is the wall time of the DMRG optimization at the accepted bond dimension, summed over all completed sweeps. It excludes electronic-structure calculations, symmetry search, Fiedler ordering, MPO construction, initial-MPS construction, and DMRG calculations at other bond dimensions. Per-sweep and complete search timings are retained in the saved JSON and CSV records.

## 2.7 Software and execution environment

The calculations used Python 3.13.5, NumPy 2.2.5, SciPy 1.15.3, PySCF 2.14.0, OpenFermion 1.7.1, OpenFermion-PySCF 0.5, and Block2/pyblock2 0.5.3. DMRG used four Block2 threads and one BLAS/LAPACK thread. The reported local timing data were obtained on an Apple M1 Pro MacBook Pro with eight CPU cores and 16 GB memory; timings should therefore be interpreted as within-machine comparisons. The benchmark does not explicitly set Block2’s internal random seed. The CISD warm start and all search tie-breaking are deterministic,

but exact runtime and convergence near the chemical-accuracy boundary can still depend on the software build, threading, and internal random-number state.

### 3 Results

Tables 3 and 4 report the smallest accepted MPS bond dimension and the corresponding DMRG optimization time. Boldface marks the smallest bond dimension within each Hamiltonian representation for a given molecule; ties are all shown in bold.

Table 3:  $\text{N}_2$  at  $R = 1.4 \text{ \AA}$ . Each entry is  $\chi_{\min}$  [DMRG wall time in seconds].

| Orbital basis              | Ordering | Fermionic SU(2)  | Raw qubit        | HCT             | Beam            |
|----------------------------|----------|------------------|------------------|-----------------|-----------------|
| Canonical                  | Standard | 23 [0.21]        | 39 [11.49]       | 12 [1.21]       | 10 [2.69]       |
|                            | Fiedler  | 31 [0.42]        | 32 [16.37]       | 13 [1.89]       | <b>8 [1.81]</b> |
| Split PM                   | Standard | 23 [0.36]        | 50 [13.19]       | 27 [19.52]      | 22 [169.73]     |
|                            | Fiedler  | <b>12 [0.47]</b> | 33 [17.46]       | 21 [8.03]       | 19 [40.65]      |
| Split PM, irrep-restricted | Standard | 23 [0.23]        | 38 [14.60]       | 10 [1.78]       | 10 [1.76]       |
|                            | Fiedler  | 33 [0.37]        | 31 [5.90]        | 12 [1.76]       | <b>8 [1.85]</b> |
| Natural                    | Standard | 24 [0.22]        | 35 [7.33]        | <b>9 [1.33]</b> | 10 [1.91]       |
|                            | Fiedler  | 33 [0.37]        | <b>28 [8.32]</b> | 10 [1.92]       | 11 [1.51]       |

Table 4: Linear  $\text{H}_8$  at adjacent H–H separation  $2.0 \text{ \AA}$ . Each entry is  $\chi_{\min}$  [DMRG wall time in seconds]. A dagger marks a run that reached the sweep-energy tolerance but not chemical accuracy by  $\chi = 300$ ; its stationary higher-energy solution is consistent with optimization trapping and is not interpreted as a bond-dimension lower bound.

| Orbital basis              | Ordering | Fermionic SU(2) | Raw qubit         | HCT              | Beam             |
|----------------------------|----------|-----------------|-------------------|------------------|------------------|
| Canonical                  | Standard | 35 [0.37]       | 123 [49.66]       | — <sup>†</sup>   | 47 [9.45]        |
|                            | Fiedler  | 35 [0.40]       | 122 [49.05]       | 54 [11.56]       | 49 [10.71]       |
| Split PM                   | Standard | 32 [0.64]       | 117 [54.96]       | 37 [12.16]       | 32 [16.73]       |
|                            | Fiedler  | <b>8 [0.23]</b> | <b>57 [18.68]</b> | <b>30 [8.63]</b> | <b>19 [6.34]</b> |
| Split PM, irrep-restricted | Standard | 35 [0.38]       | 123 [45.39]       | — <sup>†</sup>   | 45 [8.33]        |
|                            | Fiedler  | 33 [0.39]       | 105 [38.36]       | — <sup>†</sup>   | 46 [9.50]        |
| Natural                    | Standard | 35 [0.34]       | 123 [48.96]       | 52 [11.35]       | — <sup>†</sup>   |
|                            | Fiedler  | 35 [0.44]       | 121 [85.95]       | 51 [10.19]       | 48 [7.32]        |

#### 3.1 $\text{N}_2$ : Clifford compression without a localization advantage

For  $\text{N}_2$ , both Clifford constructions substantially reduce the qubit-MPS bond dimension. In canonical orbitals, the raw-qubit threshold is 39, compared with 12 for HCT and 10 for Beam. Beam with Fiedler ordering reaches  $\chi = 8$ , as does Beam in the irrep-restricted split-PM basis; the lowest HCT threshold is  $\chi = 9$  in standard-order natural orbitals. The CISD natural orbitals also give the smallest raw-qubit threshold, 28, after Fiedler ordering.

Unrestricted split-PM localization is unfavorable for the transformed  $\text{N}_2$  calculations. Relative to canonical orbitals, its HCT and Beam costs increase by approximately 32–33%, while the standard-order thresholds increase from 12 to 27 and from 10 to 22, respectively. The corresponding Pauli MPOs also grow sharply: the standard-order Beam MPO increases from bond dimension 832 to 6097. Consequently, its accepted  $\chi = 22$  calculation requires 169.73 s, despite having a smaller state bond dimension than the raw-qubit calculation. Restricting localization to occupied/virtual irrep blocks largely restores the canonical symmetry costs, MPO dimensions, and DMRG thresholds.

Fiedler ordering is not uniformly beneficial for this compact diatomic. It improves the canonical raw-qubit and Beam thresholds and strongly improves the split-PM fermionic result, but it increases several fermionic, HCT, or Beam thresholds in the canonical, irrep-restricted, and natural bases. Orbital localization and ordering must therefore be evaluated together rather than treated as independently favorable transformations.

### 3.2 H<sub>8</sub>: cooperative localization and ordering

The extended H<sub>8</sub> chain shows the expected real-space localization benefit. Split-PM localization lowers the HCT and Beam symmetry costs from 6.224 and 3.409 in canonical orbitals to 1.253 and 0.617. Combining this basis with Fiedler ordering gives the smallest threshold in every representation:  $\chi = 8$  for fermionic SU(2), 57 for raw qubits, 30 for HCT, and 19 for Beam. Thus Beam reduces the required qubit-MPS bond dimension by a factor of three relative to the raw-qubit calculation in the same orbital basis and ordering.

The irrep-restricted localization yields much smaller improvements. This contrast indicates that mixing canonical point-group blocks is important for forming orbitals localized along the hydrogen chain, whereas the same mixing is detrimental to N<sub>2</sub>. Natural orbitals also provide little reduction for raw H<sub>8</sub>: the standard thresholds remain 123 in the canonical, irrep-restricted, and natural bases.

Four transformed H<sub>8</sub> calculations satisfy the sweep-energy stopping criterion but fail Eq. (3): standard-order canonical HCT, both irrep-restricted HCT orderings, and standard-order natural Beam. At  $\chi = 300$ , these calculations remain near  $-3.6711575235$  Ha, approximately 125.5 mHa above FCI, and increasing the nominal bond cap does not lower the energy. Standard-order canonical HCT and natural Beam both reach chemical accuracy after Fiedler reordering. These observations identify local DMRG optimization behavior, rather than an insufficient representational bond dimension, as the likely origin of at least those two failures.

### 3.3 Operator complexity and runtime

Table 5 shows that Clifford compression of the state can increase the MPO bond dimension. The effect is particularly large for unrestricted split-PM N<sub>2</sub>, where the Beam MPO reaches 6097. For split-PM H<sub>8</sub> with Fiedler ordering, Beam reduces the state threshold from 57 to 19 while increasing the MPO bond dimension from 697 to 2200. That tradeoff is still favorable for the accepted DMRG calculation: the Beam calculation takes 6.34 s, compared with 18.68 s for raw qubits. For N<sub>2</sub>, however, the much larger Beam MPO dominates the smaller state bond dimension and increases the runtime.

Table 5: Maximum MPO bond dimensions. Each entry gives standard/Fiedler ordering. The fermionic and qubit MPO bond dimensions refer to different local site spaces and should be compared primarily within a representation.

| System         | Orbital basis              | Fermionic SU(2) | Raw qubit | HCT       | Beam      |
|----------------|----------------------------|-----------------|-----------|-----------|-----------|
| N <sub>2</sub> | Canonical                  | 63/87           | 291/575   | 539/861   | 832/730   |
|                | Split PM                   | 123/123         | 643/3752  | 3046/2561 | 6097/4178 |
|                | Split PM, irrep-restricted | 63/87           | 291/544   | 525/923   | 824/707   |
|                | Natural                    | 63/87           | 291/586   | 655/979   | 824/1150  |
| H <sub>8</sub> | Canonical                  | 83/83           | 349/690   | 2095/1855 | 1173/1226 |
|                | Split PM                   | 83/83           | 411/697   | 2069/1903 | 3355/2200 |
|                | Split PM, irrep-restricted | 83/83           | 349/846   | 1227/1532 | 1491/1382 |
|                | Natural                    | 83/83           | 349/693   | 1678/1496 | 1189/1154 |

The fermionic SU(2) implementation is the fastest method throughout the successful runs. At the accepted bond dimension, fermionic optimization takes 0.21–0.47 s for N<sub>2</sub> and 0.23–0.64 s for H<sub>8</sub>. Successful Beam calculations take 1.51–169.73 s for N<sub>2</sub> and 6.34–16.73 s for H<sub>8</sub>. This advantage reflects the specialized quantum-chemistry MPO, spatial-site representation, and SU(2) block sparsity. The Clifford benchmarks therefore demonstrate reduced qubit-MPS entanglement requirements, not a wall-time advantage over spin-adapted fermionic DMRG.

## 4 Limitations and next steps

The benchmark establishes trends for two STO-3G systems and a single CISD warm start per frame. The integer bisection assumes that DMRG acceptance is monotone with the allowed bond dimension; the implementation detects nonmonotone evaluated outcomes, but independent random seeds were not used to quantify optimizer variability near the threshold. The stationary H<sub>8</sub> failures show that sweep-energy convergence alone does not establish ground-state convergence. Future comparisons should repeat marginal and failed frames with controlled seeds or alternative noise schedules.

The current Pauli terms are supplied to the MPO builder in streamed Jordan–Wigner order. Sorting by coefficient magnitude, spatial support, or an MPO-specific graph heuristic may reduce intermediate and final MPO bond dimensions without changing the Hamiltonian and is a direct target for follow-up tests. Additional directions are orbital optimization against an entanglement- or MPO-aware objective, systematic Beam-pool

and beam-width studies, iterative searches restricted to Pauli-Z quasi-symmetries, and fermionic Gaussian transformations defined directly in Majorana space.

## 5 Reproduction and saved data

The authoritative driver is `scripts/benchmark_sto3g_orbital_dmrg.py`. The following commands specify the two calculations; all remaining parameters are the defaults reported above.

```
python -u scripts/benchmark_sto3g_orbital_dmrg.py \  
--system N2_corr --orderings standard fiedler \  
--min-bond-dim 1 --max-bond-dim 200
```

```
python -u scripts/benchmark_sto3g_orbital_dmrg.py \  
--system H8_corr --orderings standard fiedler \  
--min-bond-dim 1 --max-bond-dim 300
```

The complete result directories are

```
saved/results/n2_corr_sto3g_orbital_dmrg/  
saved/results/h8_corr_sto3g_orbital_dmrg/
```

Each directory contains the exact configuration in `settings.json`, aggregate thresholds in `dmrg_summary.csv`, per-bond energies and timings in `dmrg_curves.csv` and JSON form, and one subdirectory per orbital basis. The basis subdirectories retain molecular integrals and orbitals, sparse CISD states, selected Pauli generators and their individual costs, Clifford/Fiedler metadata, MPO bond dimensions, per-sweep energies and runtimes, and native Block2 MPO/MPS restart artifacts. Results are checkpointed after every completed bond-dimension calculation, and `-resume` continues an interrupted calculation only when its result-defining settings agree with the saved configuration.